AI-Powered Design Change Impact & Cost Estimation System

Enterprise Architecture Reference — Shipbuilding / Discrete Manufacturing

Source: MDL Problem Statements PS-1 (AI-Based Impact Analysis) and PS-2 (AI Tool for BOM/Class Rule Optimisation)

Problem synthesis (from source documents)

PS-1 identifies four root causes of design change failure at MDL:

1. Unclear cross-system impact visibility (structure, propulsion, weapons, electrical, outfitting)
2. Delayed and uncertain cost estimates — 15–35% cost escalation risk (MDL internal project data)
3. Schedule overruns from late assessment
4. Knowledge locked in individuals, not systems

PS-2 adds a compliance and BOM synchronisation gap:

- 25% of changes occur post-contract design finalisation (MDL commercial project data)
- Downstream impacts: structural re-analysis, weight redistribution, re-qualification testing
- Need to validate against Class rules (LR, DNV, BV) and sync 3D models with BOM

Proposed solution: DCIA — Design Change Intelligence Assistant

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Layer 1: React.js front end

Technology: React 18 + TypeScript · Tailwind CSS · D3.js for graph visualisation · React Flow for dependency maps

Key screens:

- **Natural language chat panel** — user submits change requests in plain English; system responds conversationally with impacts, costs, and next actions

- **Change submission form** — structured input for system, component, change type, magnitude
- **Impact heat-map** — colour-coded ship system diagram showing affected disciplines and severity
- **Cost breakdown dashboard** — parametric cost bands with historical comparables (recharts)
- **Compliance status panel** — Class rule flags (pass/fail/query) with LR/DNV article references
- **Approval workflow** — route change requests for sign-off with full audit trail
- **BOM sync viewer** — 3D model ↔ BOM delta display; highlights unresolved discrepancies

Design principle: Every response from the AI is traceable — sources, confidence levels, and data lineage are always surfaced to the user.

Layer 2: Enterprise AI orchestration

LLM options: Anthropic Claude (claude-sonnet) or GPT-4o · deployed on private cloud / Azure OpenAI for data residency

Components:

Component	Role
Intent classifier	Identifies change type, affected system, query intent (impact / cost / compliance / BOM)
Entity extractor	Pulls system name, component ID, change magnitude, drawing numbers from natural language
RAG pipeline	Retrieves relevant change history, standards documents, and cost records via vector similarity
Multi-agent router	Dispatches sub-tasks to specialist engines (impact, cost, compliance, schedule)
Synthesis agent	Aggregates results into a coherent, cited natural language response
Confidence scorer	Attaches confidence bands to every estimate based on data completeness

Vector store: Pinecone or pgvector (PostgreSQL) — embeds: change records, technical reports, class rule clauses, lessons learned

RAG data sources: Historical change orders (MDL) · Class society rule documents · Supplier data sheets · Project post-mortems

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Layer 3: Neo4j knowledge graph

Why Neo4j: Design change propagation is inherently a graph problem — impacts traverse relationships, not rows. Cypher queries traverse multi-hop discipline dependencies in milliseconds.

Node types:

Node label	Key properties
`DesignChange`	id, description, status, initiatedBy, date
`ShipSystem`	name, discipline (structure/propulsion/electrical/weapons/outfitting), drawing_ref
`Component`	partNumber, description, weight, material, cost_usd
`BOMItem`	quantity, unit, revision, 3d_model_ref
`ClassRule`	ruleId, society (LR/DNV/BV), description, clause, applicability
`CostRecord`	amount, currency, category (labour/material/test), year, confidence
`Supplier`	name, leadTimeDays, country, approvedStatus

Relationship types:

Relationship	Meaning
`(:DesignChange)-[:IMPACTS]->(:ShipSystem)`	Direct first-order impact
`(:ShipSystem)-[:CASCADES_TO]->(:ShipSystem)`	Cross-discipline propagation path
`(:Component)-[:PART_OF]->(:BOMItem)`	BOM membership
`(:ShipSystem)-[:GOVERNED_BY]->(:ClassRule)`	Compliance obligation
`(:DesignChange)-[:SIMILAR_TO]->(:DesignChange)`	Historical analogue (cosine similarity)
`(:Component)-[:SOURCED_FROM]->(:Supplier)`	Supply chain link
`(:DesignChange)-[:GENERATES]->(:CostRecord)`	Estimated or actual cost

Sample Cypher query — find all systems affected by a propulsion change:

```cypher
MATCH (dc:DesignChange {id: $changeId})-[:IMPACTS]->(s:ShipSystem)
      -[:CASCADES_TO*1..3]->(affected:ShipSystem)
RETURN affected.name, affected.discipline, length(path) as hops
ORDER BY hops ASC
```

Graph maintenance: Automated ingestion pipelines from PLM (Teamcenter/Windchill), ERP (SAP), and CAD exports keep the graph current. Every approved change order updates relevant nodes and edges.

Layer 4: Data & integration layer

Source system	Integration method	Data provided
PLM (Teamcenter / Windchill)	REST API / file export	Drawing revisions, BOM structure, component attributes
ERP (SAP S/4HANA)	OData API	Cost actuals, supplier lead times, procurement records
CAD / 3D model store	STEP/IFC export + parser	Geometry attributes, weight, spatial relationships

Source system	Integration method	Data provided
Classification society portals	LR MARS / DNV Veritas API	Rule updates, approval status, survey records
Document management (EDMS)	Full-text index + vector embedding	Technical reports, change orders, NCRs, lessons learned
Schedule (Primavera P6)	XML export	Critical path, float, milestone dates

Data pipeline: Apache Kafka for event streaming · Apache Airflow for batch ETL · Delta Lake for historical cost analytics

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Layer 5: Specialist intelligence engines

5a. Impact propagation engine

- Executes graph traversal (Neo4j) to find all affected disciplines
- Assigns severity scores (high / medium / low) using trained classifier on historical change data
- Outputs: discipline impact matrix, cascade depth, estimated re-work scope

5b. Parametric cost estimation engine

- Regression models trained on MDL historical change cost data (addressing PS-1: 15–35% overrun risk)
- Input features: change type, system, magnitude, affected discipline count, ship class
- Output: cost band (P10/P50/P90) with confidence score
- Fallback: analogous cost retrieval from Neo4j `CostRecord` nodes via RAG

5c. Compliance checker

- Rule graph in Neo4j maps every ship system to applicable class rules
- LLM-assisted interpretation of rule clauses for ambiguous cases
- Outputs: pass / fail / requires-survey flag per affected system, with specific rule citations
- References: LR Rules for Ships, DNV-RU-SHIP, BV NR467

5d. Schedule impact analyser

- Reads critical path from Primavera P6 integration
- Models delay propagation using activity dependency graph
- Outputs: days-at-risk per milestone, recommended sequencing adjustments

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Layer 6: Infrastructure & security

Deployment: On-premises or private cloud (Azure / AWS GovCloud) to meet shipyard data classification requirements

Security controls:

- Role-based access control (RBAC): design engineers, cost engineers, project managers, approvers
- All LLM calls routed through enterprise API gateway — no data leaves the private network
- Full audit log: every AI-generated output is logged with input, model version, timestamp, user
- Explainability: every cost estimate and impact flag includes data sources and reasoning chain

Reliability:

- Neo4j Causal Cluster (3-node) for HA
- Vector DB replicated across availability zones
- LLM fallback chain: primary model → secondary model → rule-based fallback
- Human-in-the-loop gates for changes above a defined cost or risk threshold

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Addressing the specific PS requirements

PS requirement	How DCIA addresses it
Analyse design change at early stage (PS-1)	NL interface allows submission at concept stage; graph traversal works on partial data
Predict cross-discipline propagation (PS-1)	Neo4j cascade relationships + trained impact classifier

PS requirement	How DCIA addresses it
Cost/schedule with reasonable confidence (PS-1)	Parametric model with P10/P50/P90 bands; always shows confidence score
Management decision support (PS-1)	Executive summary view; approval workflow; risk-ranked action list
Class rule validation (PS-2)	Compliance engine with real-time LR/DNV/BV rule graph
3D model ↔ BOM synchronisation (PS-2)	BOM sync viewer; delta detection on CAD/PLM import; discrepancy flagging
Post-contract change management (25% of changes, PS-2)	Full change lifecycle tracking in Neo4j; re-qualification tracking nodes

Key data quality caveat

The system's accuracy is directly dependent on the completeness and currency of historical change data in MDL's PLM and ERP systems. A data remediation programme to back-populate cost actuals and change outcomes is recommended as a pre-condition for deploying the cost estimation engine at production confidence levels.